

Customer Shopping Behavior Analysis

3,900 transactions · 18 features · 4 categories · Python · SQL · Power BI



Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Values

Review Rating only

\$59.76

Avg Purchase

Customer Data

Age · Gender · Location · Subscription Status

Purchase Data

Item · Category · Amount · Season · Discount · Shipping

Data Preparation in Python

01

Load & Explore

`pandas · df.info() · df.describe()`

02

Clean

`Imputed Review Rating by category median · snake_case rename`

03

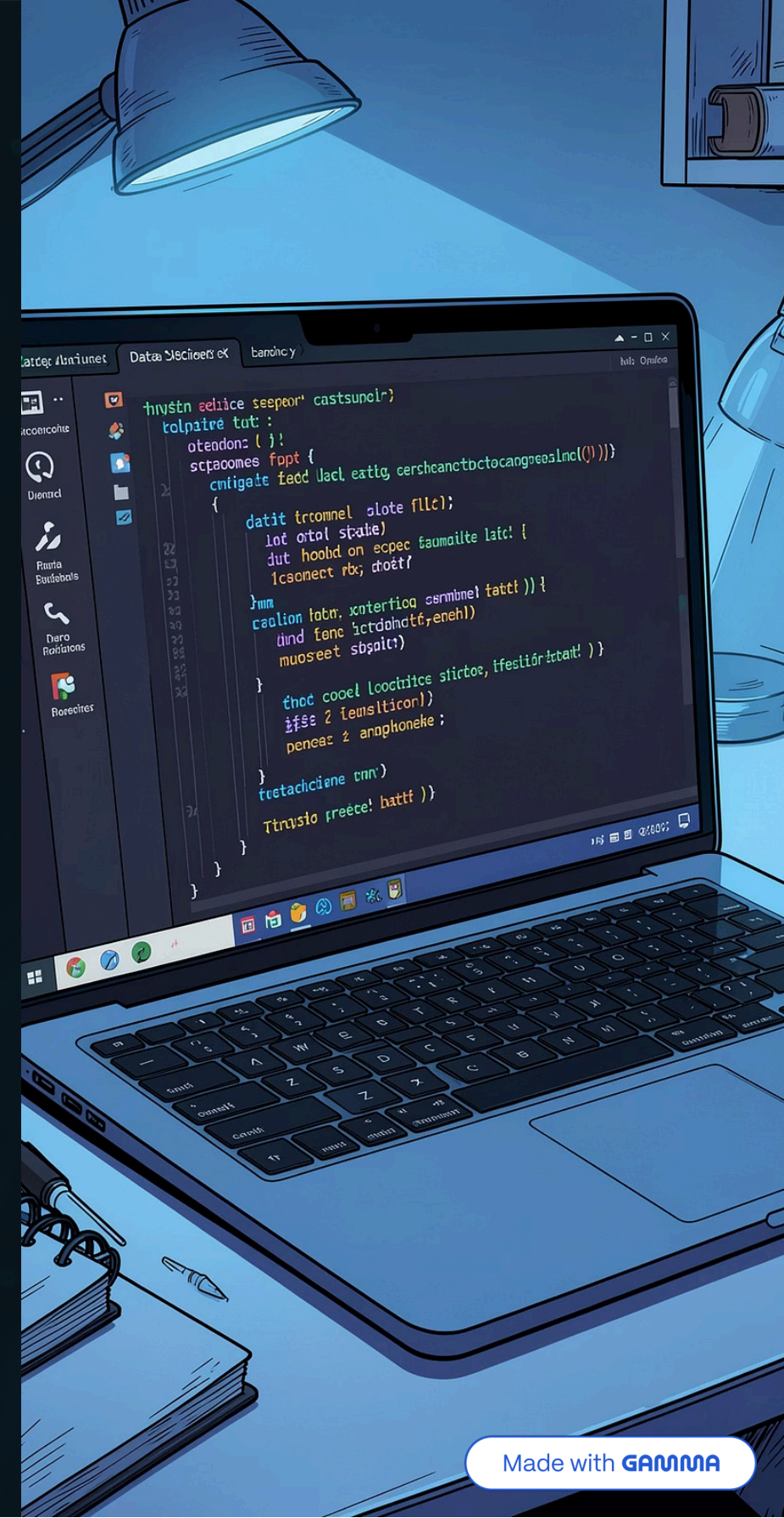
Engineer

`age_group · purchase_frequency_days · dropped redundant
promo_code_used`

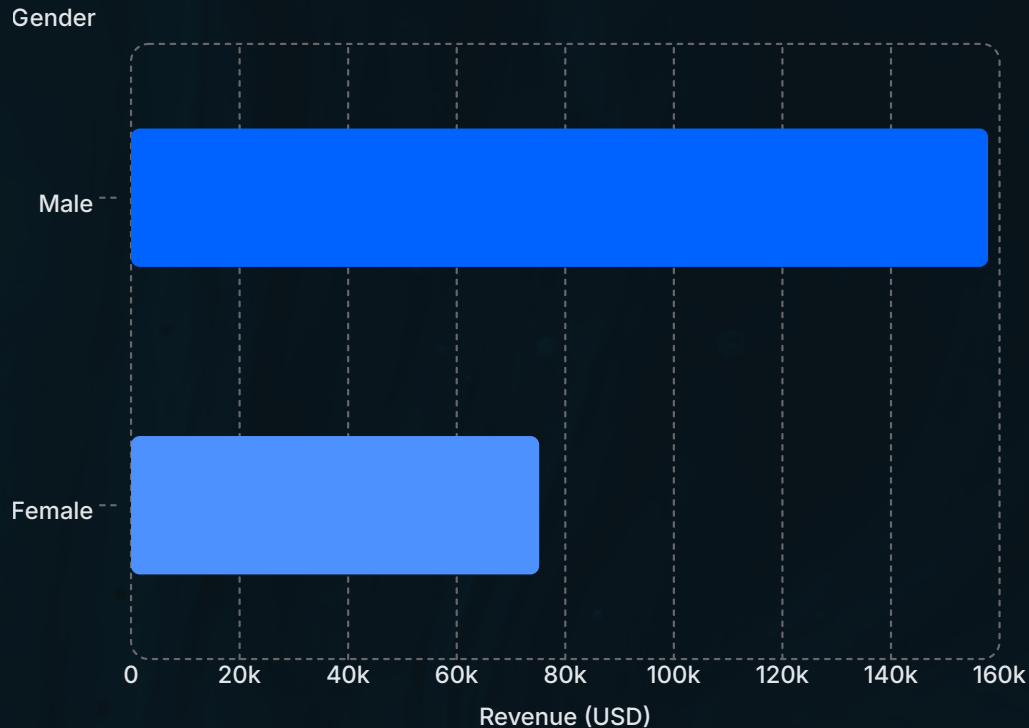
04

Load

`Cleaned DataFrame → PostgreSQL`



Revenue & Spending Insights



Revenue by Gender

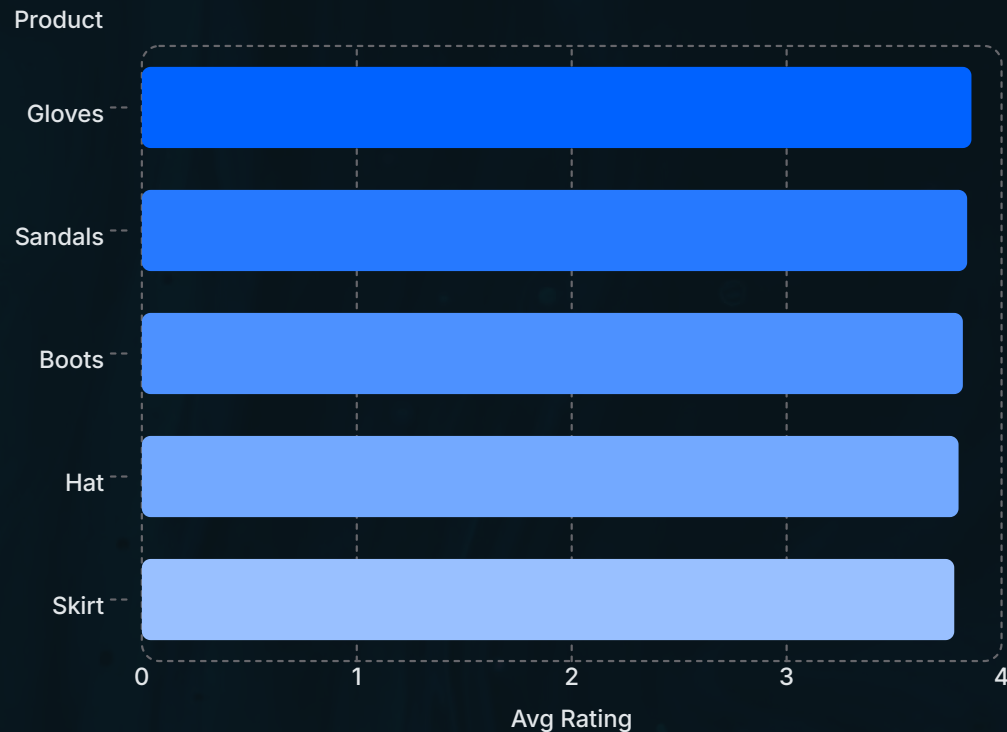
Males generate 2× more revenue than females.

High-Spending Discount Users

839 customers used discounts yet spent above average.

Products & Shipping

Top 5 Products by Rating



Shipping Type Avg Spend

Standard

\$58.46

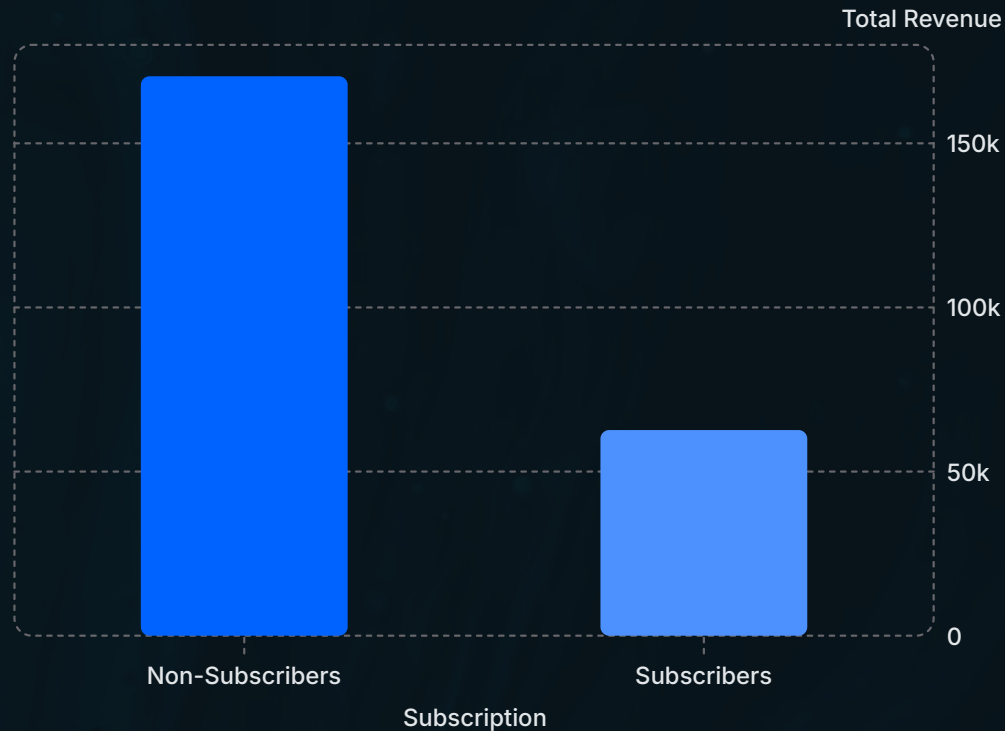
Express

\$60.48

Most Discount-Dependent

Hat · Sneakers · Coat · Sweater · Pants

Subscribers vs. Non-Subscribers



Key Findings

Avg spend nearly identical: \$59.87 vs \$59.49

Repeat buyers: 2,518 non-subscribers vs 958 subscribers

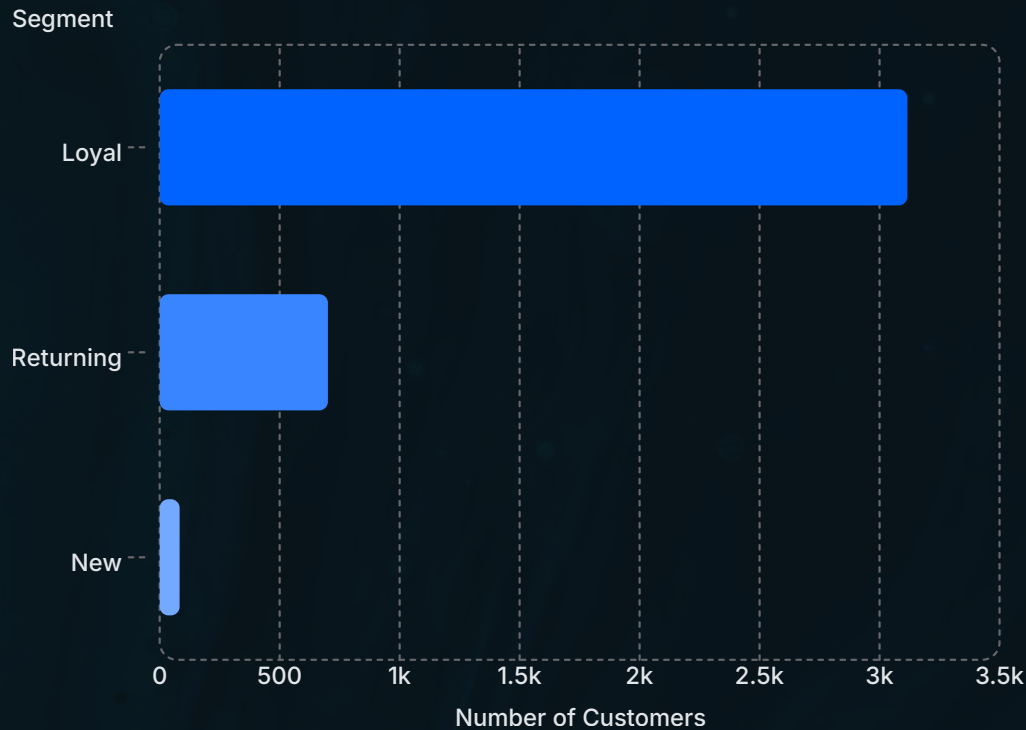
73%

Non-subscribers

27%

Subscribers

Customer Segmentation



Segment Breakdown

Loyal — 3,116

80% of all customers

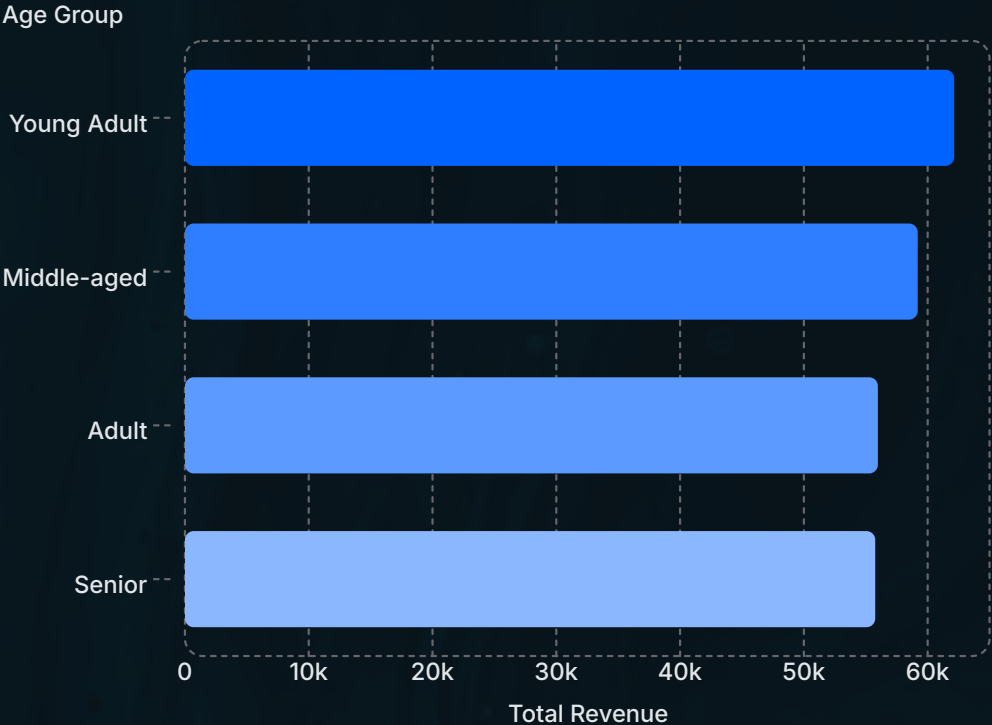
Returning — 701

18% of base

New — 83

2% — growth opportunity

Revenue by Age Group & Top Products



Top Products by Category

Category	Top Items
Accessories	Jewelry, Sunglasses, Belt
Clothing	Blouse, Pants, Shirt
Footwear	Sandals, Shoes, Sneakers
Outerwear	Jacket, Coat



Power BI Dashboard



Filters

Subscription · Gender · Category · Shipping



Visuals

Revenue & Sales by Category & Age Group



Subscription Split

27% subscribers vs 73% non-subscribers

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → Loyal segment



Review Discounts

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express shippers